

Brady Bangasser

Site Reliability & Platform Engineer

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Systems engineer focused on reliability and platform work: running production HPC and cloud infrastructure, automating operations, and keeping large-scale systems available and observable under real load.

EXPERIENCE

System Administrator · Iowa State University

Apr 2025 – Present

Ames, IA

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.
- Automated node provisioning and recovery with Ansible so failed or new nodes rejoin service quickly.
- Built Prometheus-based monitoring and alerting to catch cluster issues before users notice.
- Validated cluster performance and hardware integrity with distributed LINPACK and NCCL benchmarks, stress-testing multi-GPU InfiniBand bandwidth.
- Improved researcher accessibility by supporting user applications through OpenOnDemand and streamlining common workflows.

Head of Networking · Cardinal Space Mining

Aug 2024 – Present

Ames, IA

- Designed, built, and tested the internal network for a competition mining robot, ensuring redundant control and telemetry throughout each arena run.
- Collected multiple camera streams and relayed them to a third-party platform for live judging.
- Hardened networking equipment against wireless interference, dust, and physical damage in the competition arena.

Cloud Security Engineer · John Deere

May 2025 – Aug 2025

Johnston, IA

- Built and deployed tooling that verified 100,000+ AWS and Azure resources against company policy, NIST standards, and platform best practices.
- Designed and managed least-privilege IAM policies in AWS to tighten cloud access control.
- Wrote automated remediation for common misconfigurations flagged across the cloud estate.
- Led a migration of IAM, security, and dev-management tooling from Python to Go, cutting runtime cost by over 10%.
- Deployed high-throughput Go services on AWS EKS to run compliance checks across the cloud estate.

Research Assistant — High Performance Computing · Bethel University

Aug 2023 – Aug 2024

Saint Paul, MN

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Developed a self-healing IoT sensor mesh network to detect earthquakes, landslides, and flash floods in remote regions.

PROJECTS

Erid — Unified Cloud Images

2026

- Single Packer + Terraform pipeline that builds identical, reproducible golden images across multi-cloud and on-prem targets, applying the same hardening everywhere to eliminate drift.
- Standardized a hardening baseline applied identically to every target image, so security posture no longer varies by platform.
- Replaced per-cloud manual image setup with one definition, cutting drift and setup time across environments.

Tech: Terraform, Packer, AWS, Azure, GCP, Linux

Omni HTTP Router

2025–2026

- Wrote a compiler that lowers a routing DSL directly to native object files, and compiles matching Terraform and container deployments alongside the app binary.
- Benchmarked the static router against conventional middleware stacks to quantify the latency and allocation savings.

Tech: Rust, LLVM, Terraform

Computer Vision Parking Enforcement Radar

2025–2026

- Architected a Redis Streams pipeline for low-latency, high-throughput image and detection routing from a REST API to processing servers, backed by PostgreSQL geospatial storage.
- Tuned the broker for low-latency, high-throughput delivery from the REST API to processing servers.

Tech: Python, Redis Streams, PostgreSQL, REST

Public Data Scraping and Processing

2025

- Built a parallel, scalable pipeline to scrape, clean, and store public datasets, deployed via a Terraform CI/CD pipeline to AWS and a personal Kubernetes cluster.

Tech: C++, C, Python, Lua, Rust, MySQL, Docker, PyTorch, Terraform, Kubernetes

Gin Gonic Routing & Middleware Framework

2025–2026

- Wrote a Rust tool that auto-generates and protects Go (Gin) routes by file path, with middleware for permission bits, request IDs, and structured logging.
- Generated per-route middleware for permission bits, request IDs, and structured logging.

Tech: Rust, Go, Gin Gonic

EDUCATION

M.S. Computer Science · Iowa State University

Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science · Iowa State University

Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List · Minors in Cybersecurity and Data Science

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, TypeScript

Infrastructure & Platform: Kubernetes, Docker, Terraform, Packer, Ansible, SLURM, Prometheus, Apptainer, Linux/Unix, Git, CI/CD

Cloud: AWS, Azure, Google Cloud, AWS EKS

Data & Backends: PostgreSQL, Redis

Security: NIST standards, Least-privilege IAM

INTERESTS

Aviation (flying small aircraft), rock climbing